

SAISRIVATHSAN MANIKANDAN

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PROFESSIONAL SUMMARY

Computer Science graduate student specializing in deep learning, computer vision, multimodal AI, and LLM systems. Skilled in Python, C++, PyTorch, FastAPI, distributed training, on-premises inference, AWS SageMaker, and MLOps with experience implementing research models and building end-to-end AI pipelines.

EDUCATION

Arizona State University

Master of Science in Computer Science

Tempe, AZ
Expected May 2027

- **Relevant Coursework:** Cloud Computing, Data-Intensive Systems for Machine Learning, Data Mining

Arizona State University

Bachelor of Science in Computer Science

Tempe, AZ
May 2026

- **Relevant Coursework:** Data Structures & Algorithms, Image Processing & Analysis (Deep Learning), Machine Learning

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Swift

AI/ML: PyTorch, Swin Transformer, Hugging Face, Computer Vision, CNNs, Scikit-learn, Generative AI, LLMs, Distributed Data Parallel (DDP), Pandas, NumPy

Backend & Cloud: FastAPI, Uvicorn, Asyncio, REST APIs, AWS, Amazon SageMaker, Azure, Firebase, Linux

DevOps & Tools: Docker, Git/GitHub, GitHub Actions, Weights & Biases (W&B), TensorBoard, Pillow, python-pptx, OpenXML

RESEARCH PROJECTS

Medical Imaging AI System with Swin Transformers | *Python, PyTorch, Timm, TorchVision*

- Implemented and adapted state-of-the-art medical-imaging methods from recent research literature, applying teacher-student learning and uncertainty-aware pseudo-labeling to multi-label CheXpert classification across 14 pathologies, achieving **0.9004 mean AUC**.
- Built Swin Transformer-based systems for TB lesion localization with Feature Pyramid Network and Faster R-CNN (**mAP@0.5: 0.66, FROC: 0.59**) and anatomical segmentation with Swin-UNet V2 (**Dice: 0.96**), adapting published architectures for new medical-imaging tasks.
- Engineered a multi-GPU PyTorch Distributed Data Parallel (DDP) pipeline with mixed precision, optimized data loading, and automatic checkpoint resumption for fault-tolerant training.

PROJECTS

SlideSight | [GitHub](#) | [Demo](#) | *Python, FastAPI, Qwen3-VL, Gemma4, Pillow*

1st Place & Fan Favorite

- Built an on-premise multimodal accessibility pipeline for the ASU AIR Spark Challenge using Qwen3-VL-32B-Instruct, FastAPI, and python-pptx, processing **380 images across 505 slides** from 10 university course decks at approximately **1.4 seconds per image** while keeping data on university hardware.
- Engineered a confidence-gated human-in-the-loop workflow combining model confidence, slide-text overlap checks, spatial occupancy, and Pillow color-histogram analysis; automatically generated **228 alt-text descriptions**, silenced **122 decorative graphics**, and escalated **30 ambiguous images** for review.
- Implemented a deterministic **WCAG 2.1 AA** audit engine for missing titles, typography, table headers, hyperlinks, and reading order, while using bounded asynchronous concurrency and a thread-safe FastAPI job registry to flag **531 accessibility violations**.

Zesto-iOS Recipe App | [App Store](#) | *Swift, Firebase, REST API, OpenAI API*

- Built an iOS application integrating LLM-powered conversational features with streaming responses and caching layers to reduce request latency and API overhead.
- Developed an automated receipt-parsing and recommendation pipeline that normalized grocery items into standardized entities and synchronized real-time application state through Firestore and backend APIs.

CERTIFICATIONS & TRAINING

AWS Certified AI Practitioner | [Credential](#)

MLOps Masterclass (Workshop) | [Certificate](#)